

Map-Making with QGIS

In Part I of this tutorial, we looked at how to visualise a shapefile, add styles and query the attribute table. In this part, let us explore how to perform geoprocessing, plot custom data and prepare a map for publishing.

We will continue to use QGIS version 1.7.3 (<http://download.qgis.org>) and the Alaska dataset (http://download.osgeo.org/qgis/data/qgis_sample_data.zip).

Geometrical operations

QGIS has a number of tools for geoprocessing and data management, like nearest-neighbour analysis, buffers, union, polygon centroids, split vector layer, merge shapefiles, etc. Let's perform some of the most commonly used operations.

The intersection of two layers

First, let's try to find out all the airports that lie in the tundra region:

1. Open *tundra.shp* and *airports.shp* as vector layers.
2. From the menu, select **Vector > Geoprocessing Tools > Intersect**.
3. In the *Intersect* dialogue box (Figure 1), select the

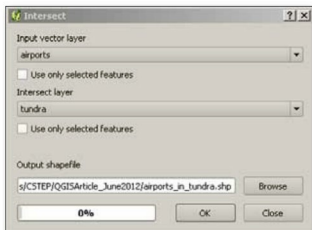


Figure 1: Intersect dialogue box



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airports as the *Input vector layer* and *tundra* as the *Intersect layer*.

4. Give a name to the output file, e.g., *airports_in_tundra*.
5. Hit OK and click Yes on the resultant dialogue box to add the output layer to the map canvas.

Polygon centroids

A layer with polygon geometry can be converted to a point layer, each point representing the centroid of the polygon:

1. Click **Vector > Geometry Tools > Polygon Centroids**.

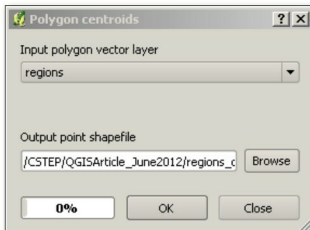


Figure 2: Polygon centroid dialogue